GreenCart

An AI-Powered Sustainable Shopping Platform for the Tampa Bay Student Community

Semester Environmental Action Project

IDH 3350: Natural Science

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Live platform: green-cart-six.vercel.app

Research blog: env-blog.vercel.app

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1. Abstract

This document chronicles the development of **GreenCart**, a full-stack web application built over the Spring 2025 semester as a practical environmental action project for IDH 3350 at the University of South Florida. GreenCart addresses the *attitude–behavior gap* in sustainable consumer behavior by giving students and Tampa Bay residents the tools to discover eco-conscious products, receive AI-generated sustainability scores for arbitrary product URLs, engage with local environmental organizations, and collectively reflect on their purchasing habits through a research-grade survey instrument. The project combines software engineering, behavioral science, and community design to lower the friction between caring about sustainability and acting on that care. This report documents the activity timeline and deliverables, describes the resulting platform and its five core modules, reflects on how the project connects to key class concepts, and offers an honest introspective account of what the process felt like and how it has changed my outlook as a consumer and a builder.

2. Introduction and Motivation

> "Consumer behavior is one of the most significant drivers of environmental change in the modern world. Every purchasing decision, travel habit, and disposal choice made by billions of people adds up to a planetary impact that science can no longer ignore."
> *Tuan Khang Phan, "Consumer Behavior Impacts on Environmental Sustainability," IDH 3350 research paper*

When I enrolled in IDH 3350, I was already studying computer science and had a growing interest in applying technology to sustainability challenges. The semester project prompt asked us to engage in an activity that creates *real positive impact* on an environment we inhabit, not a hypothetical proposal, but something we actually build, do, and live with.

For me, the most natural response was to build a platform. I had watched family and friends struggle to answer a simple question when shopping online: *"Is this product actually good for the environment, or is this just marketing?"* Greenwashing is rampant, eco-labels are inconsistent, and the cognitive effort required to cross-reference certifications, read ingredient lists, and compare carbon estimates is far beyond what the average shopper will invest at the moment of purchase.

That is the gap GreenCart is designed to close. It sits at the intersection of three circles: the behavioral science concept of the attitude–behavior gap (which we explored extensively in class), the emerging capability of large-language-model AI to synthesize complex product information, and the social fabric of a student community that actually cares about these issues but lacks an easy on-ramp to act on that care.

3. Documentation of the Activity

3.1. Project Timeline

Phase / Weeks	Activity
Weeks 1–2	Research and ideation. Read consumer behavior and sustainability literature (*attitude–behavior gap*, social norms, information asymmetry). Drafted research paper for the class.
Weeks 3–4	Technology prototyping. Stood up a Next.js 15 project, configured Clerk authentication, Supabase database and storage, and tested the Anthropic Claude API for sustainability scoring.
Weeks 5–7	Core feature development. Built the AI Analyzer pipeline, Marketplace catalog with filters, and initial product data set (~190 items).
Weeks 8–10	Community module. Designed and implemented the three-column social feed, volunteer opportunity matching, Tampa Bay organization directory, and user post creation with image upload.
Weeks 11–12	Survey instrument. Designed a 22-question, 6-section Likert survey grounded in class concepts (awareness, attitudes, behavior, social influence, barriers, platform impact).
Weeks 13–14	Guide, landing page, and publication. Wrote onboarding content, user guide (with IDH 3350 references), and deployed to Vercel. Published companion research blog at env-blog.vercel.app.
Finals week	Documentation, reflection writing, and final polish.

Table 1. Semester development timeline for GreenCart.

3.2. Technology Stack and Tools

Technology	Role in GreenCart
Next.js 15 (React 19)	Full-stack app framework; server and client components, API routes
Anthropic Claude API	AI sustainability scoring engine; generates scores, carbon estimates, concerns, strengths, and greener alternatives
Supabase	PostgreSQL database (posts, comments, products) and object storage (user-uploaded images)
Clerk	User authentication and identity management
Cheerio	Server-side HTML scraping to extract product metadata from arbitrary URLs
Vercel	Continuous deployment; live at green-cart-six.vercel.app
OpenStreetMap Nominatim	Reverse geocoding for the community Location feature
React Icons	Consistent SVG icon library throughout the UI

Table 2. Core technology stack powering GreenCart.

3.3. Deliverables and Evidence

The following deliverables document the activity. Links are clickable in the PDF.

1. **Live web platform:** green-cart-six.vercel.app
   A publicly accessible, fully deployed web application with five distinct feature modules.
2. **Research blog:** env-blog.vercel.app
   A companion blog presenting the academic research paper *"Consumer Behavior Impacts on Environmental Sustainability"* written for IDH 3350, complete with abstract, body, references, and visual layout.
3. **Source code:** available in the project repository.
   The full Next.js codebase (app router, API routes, data modules, components, styles) constitutes the primary technical artifact of this project.
4. **Survey instrument:** 22 Likert-scale questions across 6 sections, embedded at `/survey`, designed to collect real data on student sustainable consumer behavior and attitudes toward the platform.
5. **Community data seed:** A curated dataset of Tampa Bay environmental organizations, volunteer opportunities (mangrove restoration, coastal cleanups, zero-waste workshops, water quality sampling, native plant gardens), and community posts.
6. **Sustainability product catalog:** 190+ eco-labeled products across three categories (EcoLiving, EcoTech, EcoFashion) with sustainability scores, resale flags, pricing, and student feedback.

4. The Resulting Product: GreenCart Platform

GreenCart is organized into five interconnected modules. Each module addresses a distinct barrier in the journey from environmental awareness to action.

4.1. Module 1: AI Product Analyzer (`/analyze`)

The centerpiece of GreenCart is the AI Analyzer. A user pastes any product URL from any retailer; the server scrapes the product page, sends the extracted content to Claude (Anthropic's large language model), and returns a structured sustainability report containing:

- An **overall sustainability score** (0–100) rendered as an animated SVG ring
- **Category scores** across five dimensions: materials & ingredients, manufacturing processes, labor & ethics, packaging, and durability/repairability
- A **carbon estimate** in kg CO_2e with lifecycle breakdown and qualitative band (e.g., "Low," "Moderate," "High")
- **Strengths** and **concerns** listed as bullet points
- **Actionable recommendations** for the consumer
- **Greener alternatives** with names, links, estimated scores, and justification

The key design decision was to present AI scoring as *guidance*, not certification, keeping the user's critical judgment in the loop while dramatically reducing the time cost of sustainability research.

4.2. Module 2: Eco Marketplace (`/marketplace`)

The Marketplace provides a curated catalog of 190+ products tagged with sustainability scores, resale/secondhand badges, category labels (EcoLiving, EcoTech, EcoFashion), student feedback quotes, price, and star ratings. Users can:

- Filter by category, minimum rating, maximum price, or resale-only
- Switch between grid and list views
- Click "AI Score" on any listing to route directly to the Analyzer
- Submit their own product listings (authenticated users), with optional image upload to Supabase storage

The explicit **resale filter** is intentional: secondhand and refurbished goods consistently represent the highest-impact consumption choice a person can make, and surfacing them alongside new products normalizes circular economy behavior.

4.3. Module 3: Community Hub (`/community`)

The Community module is a three-column social interface modeled loosely on professional social networks but focused entirely on local environmental action in Tampa Bay. It includes:

- A **social feed** where authenticated users can post sustainability updates, share links (with auto-linked URLs), attach photos (uploaded to Supabase), tag their feeling (with an emoji

picker), and add their location (typed or auto-detected via the browser Geolocation API and OpenStreetMap reverse geocoding)
- A **volunteer opportunity matcher** that ranks local opportunities by overlap with the user's selected cause interests and availability schedule
- A curated **Tampa Bay organization directory** with links to real local groups working on coastal restoration, water quality, biodiversity, and zero-waste initiatives
- **Trending hashtags** (e.g., `#MangroveRestoration`, `#ZeroWaste`, `#TampaBayWatch`) to create a sense of shared local momentum
- **Cause interest chips** (cleanup, marine, restoration, climate, water, education, biodiversity) that personalize the volunteer feed

4.4. Module 4: Consumer Behavior Survey (`/survey`)

The survey instrument was designed to serve double duty: it provides useful data for my IDH 3350 research paper, and it prompts users to reflect on their own behavior, which is itself an intervention. The 22 questions are organized into six sections:

A. Environmental Awareness (4 items)
B. Attitudes Toward Sustainable Consumption (4 items)
C. Actual Purchasing Behavior (4 items)
D. Social & Peer Influence (3 items)
E. Barriers to Sustainable Shopping (3 items)
F. GreenCart Platform Impact (4 items)

After submission, users receive a personalized **verdict** (e.g., "Strongly Pro-Sustainability," "Eco-Conscious but Constrained," "Low Eco-Consciousness Indicated"), per-section averages visualized as colored bars, and a full per-question breakdown. Seeing their own scores laid out clearly is intended to create a moment of self-reflection that nudges behavior.

4.5. Module 5: User Guide (`/guide`)

A step-by-step onboarding page written specifically with IDH 3350 students in mind, explaining how to use each feature, what the scores mean (and don't mean), and how GreenCart connects to the course's sustainability concepts.

5. Connection to Class Concepts

5.1. The Attitude–Behavior Gap

Perhaps the most important concept from IDH 3350 for understanding GreenCart is the **attitude–behavior gap**: the well-documented tendency for people to hold pro-environmental values without translating those values into action. Research consistently shows that most consumers *care* about sustainability but are stymied by information costs, price premiums, inconvenience, and social norms that make conventional consumption the path of least resistance [**?**].

GreenCart attacks each of these friction points directly:

- **Information costs:** AI Analyzer produces a score in seconds instead of requiring hours of research.
- **Price barriers:** Marketplace surfaces budget-friendly resale options alongside new eco products.
- **Inconvenience:** The platform integrates discovery, scoring, and community in a single interface rather than requiring visits to multiple sites.
- **Social norms:** The Community feed, trending tags, and volunteer matching create visible peer activity around sustainable choices, leveraging social proof as a positive influence.

5.2. Information Asymmetry and Greenwashing

A recurring theme in class was the problem of **information asymmetry** in markets: companies know far more about the true environmental impact of their products than consumers do, and this gap is frequently exploited through greenwashing. The AI Analyzer is, at its core, an attempt to partially close this asymmetry. By applying a consistent scoring rubric across product categories and surfacing specific concerns (e.g., "Contains microplastics," "No living-wage certification visible"), it shifts the information balance toward the consumer.

The platform also acknowledges its own epistemic limits; scores are labeled as "AI guidance, not certification," which is itself an exercise in honest framing that class readings on scientific literacy and risk communication emphasize as essential for public trust.

5.3. Collective Action and Community Organizing

IDH 3350 taught us that individual behavior change, while necessary, is insufficient on its own. Sustainable futures are built through **collective action** at multiple scales: peer networks, community organizations, advocacy, and infrastructure change. The Community module of GreenCart is designed with this understanding. Volunteer matching is not merely a convenience feature; it is an attempt to lower the activation energy for someone to move from passive awareness to organized participation in Tampa Bay's environmental groups. The cause-matching algorithm prioritizes alignment between user interests (e.g., marine restoration, climate education) and opportunity type, making the translation from *interest* to *action* as frictionless as possible.

5.4. Systems Thinking and Lifecycle Awareness

The carbon estimate card in the Analyzer module is the most direct expression of **systems thinking** in GreenCart. Rather than presenting a single "eco" badge, it breaks down CO_2e by lifecycle stage: raw materials extraction, manufacturing, transportation, use phase, and end-of-life, and identifies the main impact drivers. This mirrors the lifecycle thinking we discussed in class as a prerequisite for genuine environmental literacy, as opposed to the superficial green labeling that dominates mainstream consumer culture.

5.5. Consumer Sovereignty and Agency

A philosophical premise running through the platform is that consumers are **not passive recipients** of market forces but active agents whose aggregated choices shape production systems over time. The survey's Section B (*Attitudes Toward Sustainable Consumption*) explicitly probes this belief: "I believe individual purchasing choices can make a meaningful difference to the environment." Seeing one's own score on this dimension, alongside the scores for barriers and actual behavior, creates a feedback loop between self-perception and potential agency that the platform is designed to catalyze.

6. Impact on the Environments I Engage With

6.1. The Student Community (Level 2 & 3 Scopes)

The most immediate environment GreenCart is designed to impact is the student community at USF and the broader Tampa Bay area. By providing a free, publicly accessible tool that reduces the effort required to shop sustainably, the platform has the potential to shift purchasing patterns for anyone who uses it. The ~190 products in the Marketplace represent real purchasing alternatives; the volunteer opportunities represent real organizations doing real restoration and education work; and the social feed creates a real space for community visibility around environmental choices.

Critically, the platform is also a **shared reference point**. When I posted about GreenCart in the IDH 3350 community and shared the research blog, I was contributing to a common vocabulary around the attitude–behavior gap and sustainable consumption that the class had been building all semester. A tool that makes that vocabulary actionable, turning a theoretical concept into a button you can click, bridges the gap between the academic and the lived.

6.2. The Information Environment

Beyond individual behavior, GreenCart makes a small contribution to a larger and more important project: improving the quality of sustainability information available to everyday consumers. Every analysis run through the platform produces a structured, multi-dimensional assessment that goes significantly beyond the binary eco/not-eco framing of most current labeling systems. If enough people use and trust tools like this, market pressure on producers to improve their sustainability profiles will increase, creating a systems-level feedback loop.

6.3. My Own Environment: Skills, Habits, and Dispositions

Building GreenCart has changed the way I personally shop. I now run product URLs through the Analyzer before significant purchases. I have switched several of my routine purchases to secondhand options surfaced by the Marketplace filters. More fundamentally, the act of building the survey instrument forced me to confront my own answers to those 22 questions, and several of my responses in Sections C and E (actual behavior and barriers)

were less flattering than my responses in Section B (attitudes). That gap is uncomfortable and instructive.

7. Introspective Reflection

7.1. How It Felt to Do This Project

Honestly, this project was one of the more demanding things I have undertaken as an undergraduate student, not because the individual technical components were unfamiliar, but because the goal was to build something *real*: something that would actually be used, that would reflect my own values in its design decisions, and that would be evaluated not only on technical execution but on genuine impact.

There were weeks when the gap between what I had built and what I imagined was discouraging. The AI scraping pipeline frequently ran into retailer bot-detection; the Supabase schema needed to be redesigned twice; the Community module took far longer than expected because designing a social experience that feels natural is genuinely hard. But there was also something deeply motivating about working toward a problem I actually care about. Knowing that the survey questions would produce data relevant to a paper I had written for this class gave the code a weight it would not have had if I were building a generic todo app.

One moment stands out in particular. About ten weeks in, I ran the Analyzer on a product I had purchased a month earlier and was surprised to find a sustainability score of 28 out of 100, with concerns including non-recyclable composite materials and supply chain opacity. I had bought it without thinking. That moment of confronting my own behavior through a tool I had built was strange and clarifying in equal measure. It is one thing to understand the attitude–behavior gap as a concept; it is another to see your own score.

7.2. How This Project Has Changed Me

Before this semester, I thought of sustainability as a domain issue, something for environmental science students, policy researchers, or activists. Building GreenCart has convinced me that it is equally, maybe more urgently, a **design and information problem**. The environmental outcomes we care about are downstream of millions of small decisions made by people who are not experts, who are busy, and who are navigating markets designed to obscure the information they need to decide well. Making that information legible, accessible, and actionable is work that technologists and designers are uniquely positioned to do, and it is work that matters.

I have also become more comfortable with what I would call **productive humility**: the platform does not solve greenwashing, it does not replace lifecycle assessments, and its AI scores are imperfect. But imperfect, accessible, and free is often more impactful than perfect, inaccessible, and paywalled. Naming that clearly in the platform's own copy ("AI guidance, not certification") was an important lesson in honest communication under uncertainty; something the class reading on scientific literacy and public trust made very clear.

7.3. Will I Continue?

Yes, without question. The most significant outcome of this project is not the platform itself (though I intend to keep it running and improving), but the clarity it has given me about what I want to do with my skills. I am planning to:

- Continue developing GreenCart post-semester, with a focus on making the survey data collection more robust (server-side persistence to Supabase, aggregate visualizations, shareable results).
- Expand the community module to include real-time events posted by Tampa Bay environmental organizations.
- Build additional AI analysis dimensions, including a supply chain transparency score and a circularity index.
- Share the platform actively with student organizations at USF and HCC as a practical tool for sustainability education.
- Explore whether the platform's survey data, if enough responses are collected, could become the basis of a more formal undergraduate research project.

More broadly, I intend to keep asking the question this project taught me to ask: *Where is the gap between what people care about and what they do, and how can better design and information close it?* That question is not limited to sustainability. But sustainability is where I found it first, and it is where the stakes are highest.

8. Conclusion

GreenCart began as a semester project and became, over fourteen weeks, a real tool that real people can use to make more informed choices about the products they buy and the communities they join. It is an expression of a conviction this class helped me develop: that engaged citizens do not only vote and volunteer and reduce and reuse; they also build. They build platforms and instruments and communities and knowledge systems that make sustainable living easier for the people around them.

The attitude–behavior gap is real. Information asymmetry is real. The inconvenience of sustainable consumption is real. But none of these are fixed features of the world. They are design problems, and design problems have solutions. GreenCart is one small attempt at a few of those solutions, and building it has made me a more intentional consumer, a more honest communicator, and a more committed practitioner of the sustainable futures we discussed in class.

References

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A. Platform Screenshots

The figures below are screenshots captured from GreenCart and related project outputs.

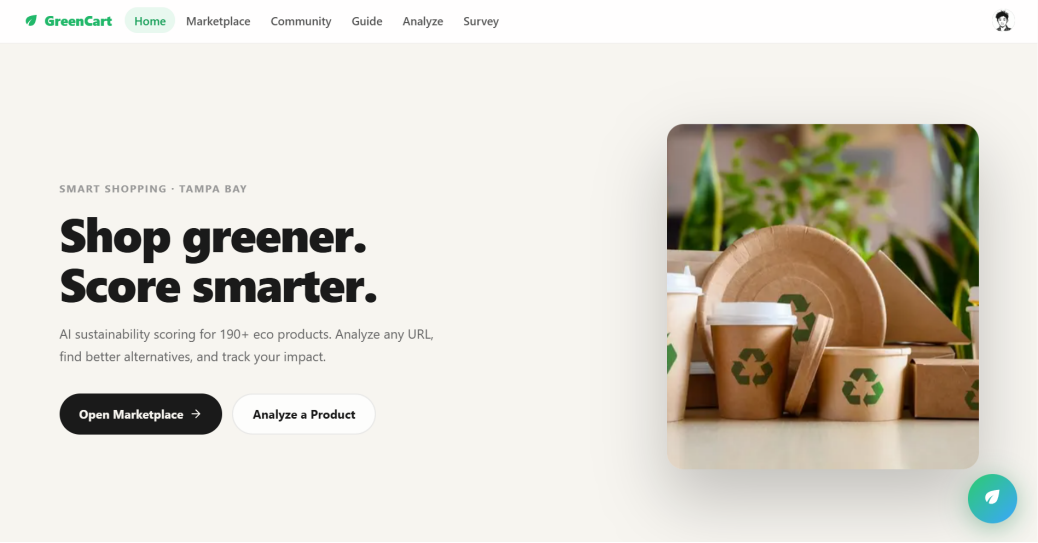


Figure 1. GreenCart landing page showing the platform's value proposition and animated sustainability impact metrics.

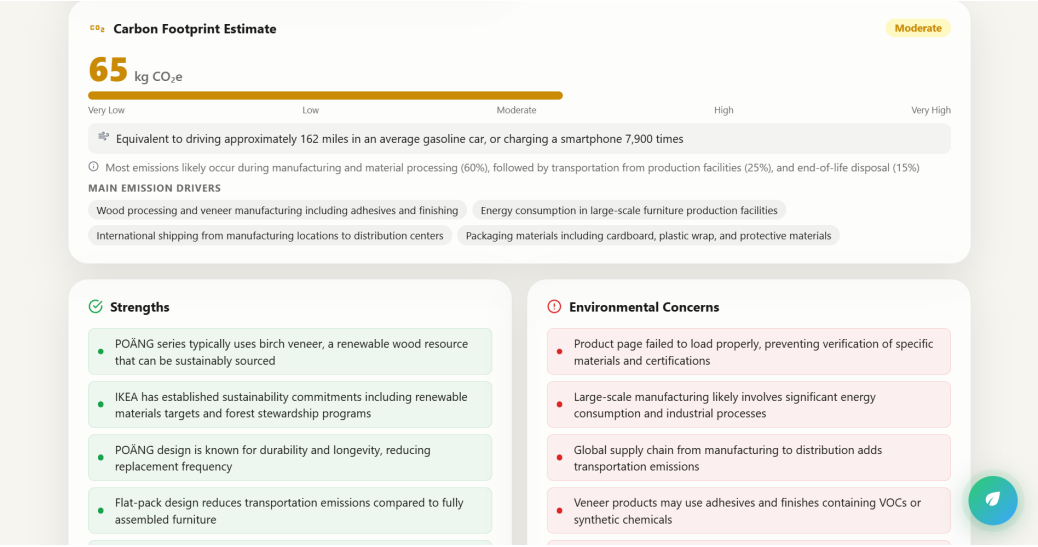


Figure 2. AI Analyzer output showing the sustainability score ring, five-dimension category scores, kg CO_2e carbon estimate with lifecycle breakdown, and recommended greener alternatives.

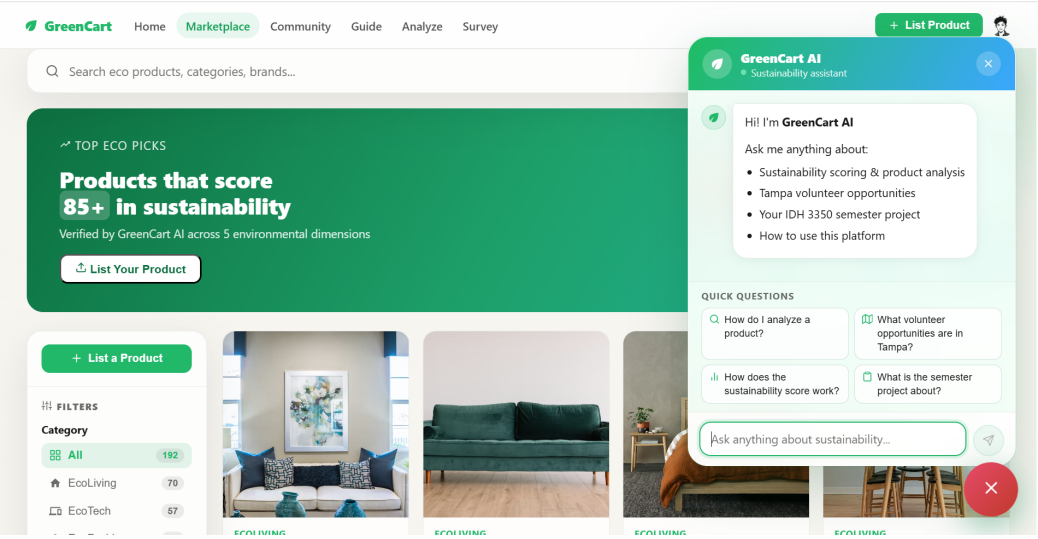


Figure 3. Eco Marketplace catalog with sustainability score pills, resale/secondhand badges, category filters, and direct links to the AI Analyzer.

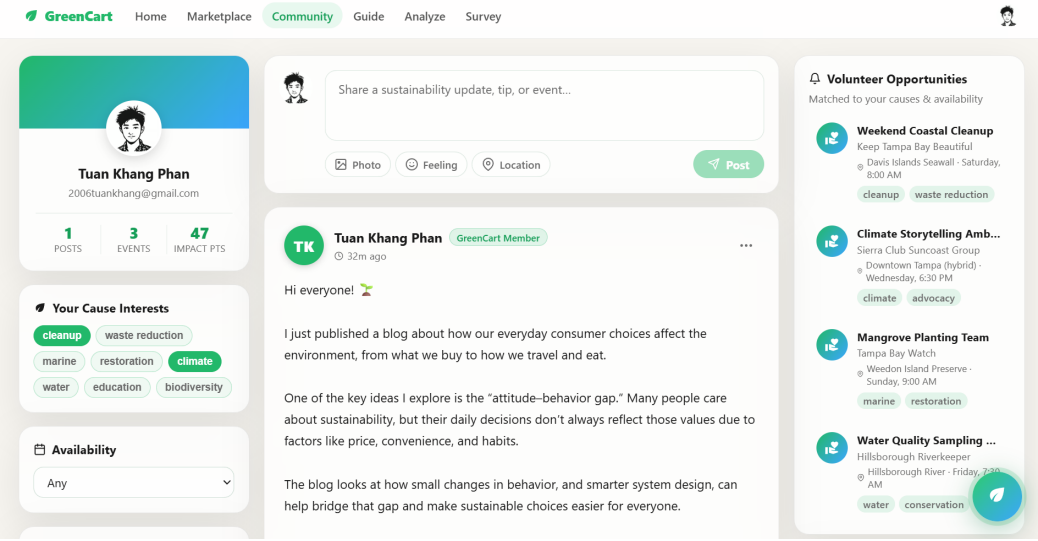


Figure 4. Community Hub with the social post composer (showing Feeling picker and Location features), cause-matched volunteer opportunities, and local Tampa Bay organization directory.

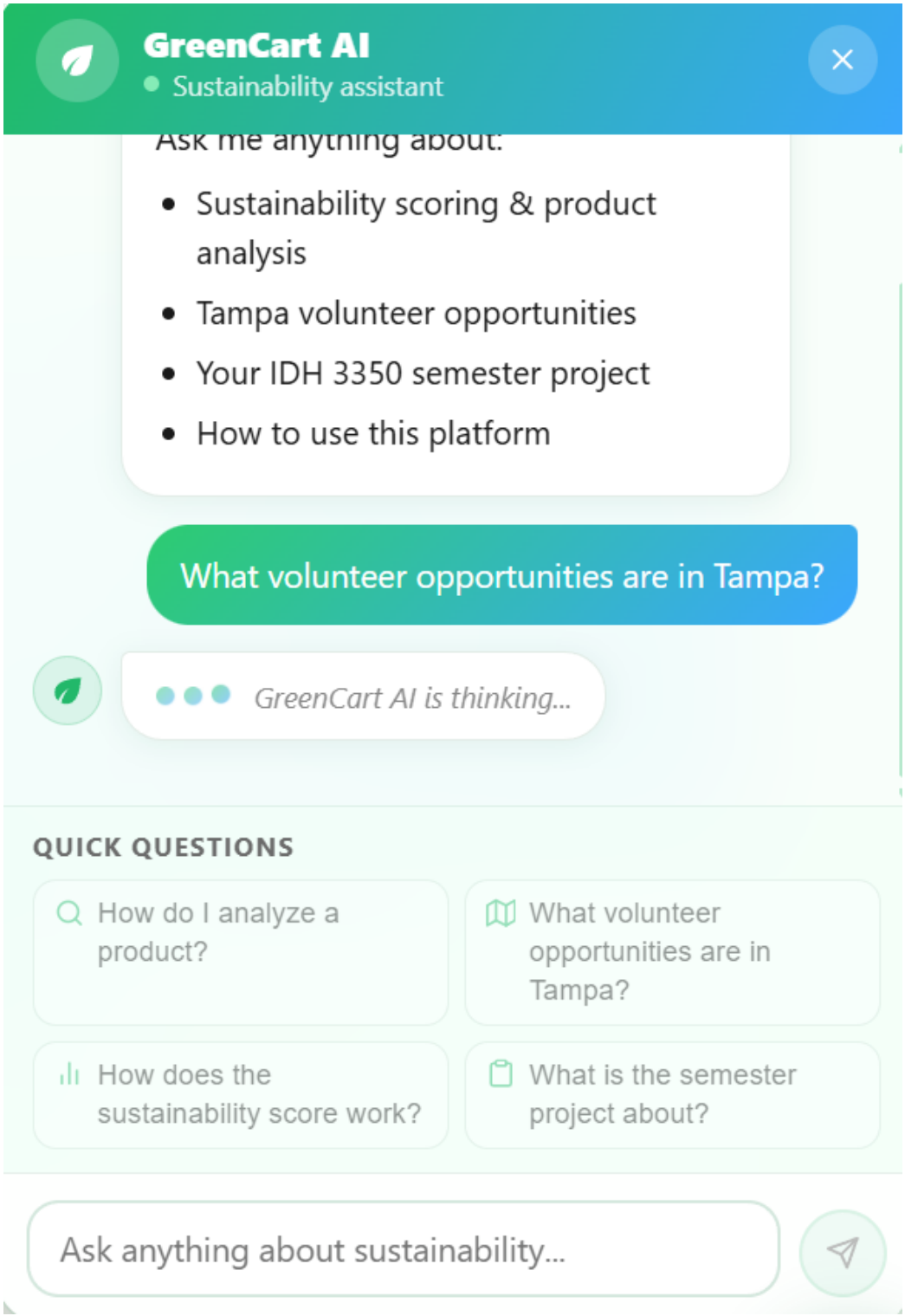


Figure 5. GreenCart AI chatbot assistant interface used for quick sustainability guidance, product-related questions, and follow-up learning prompts.

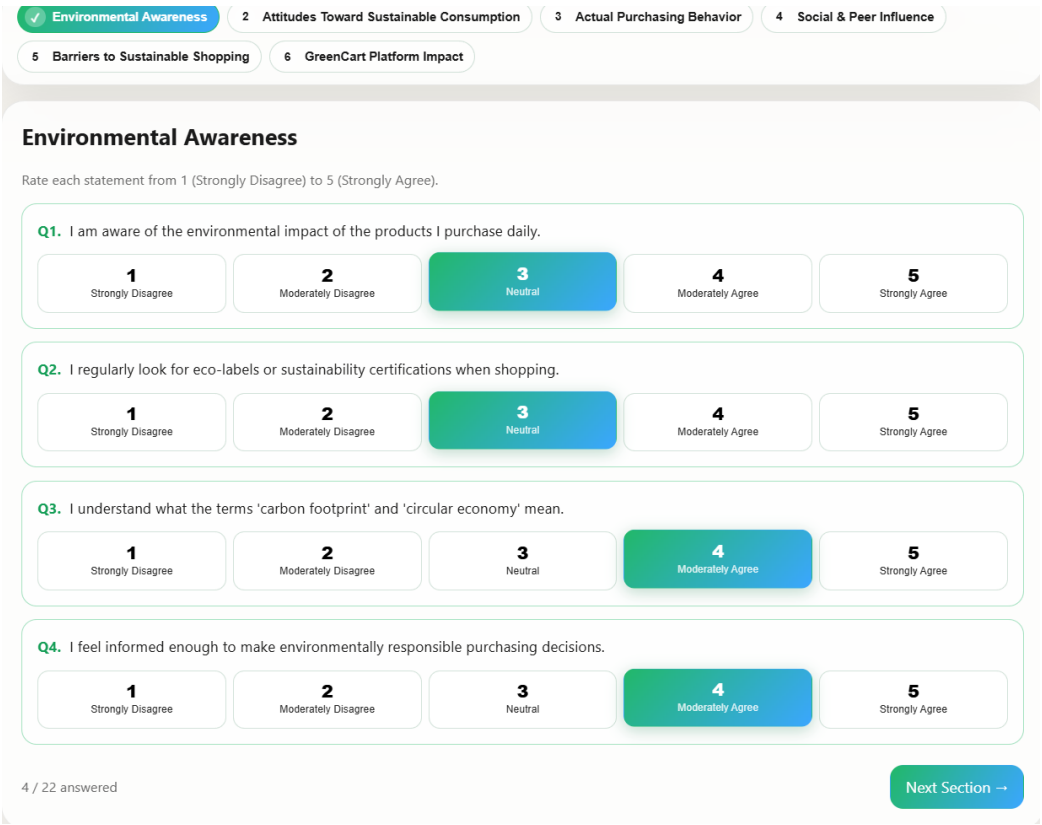
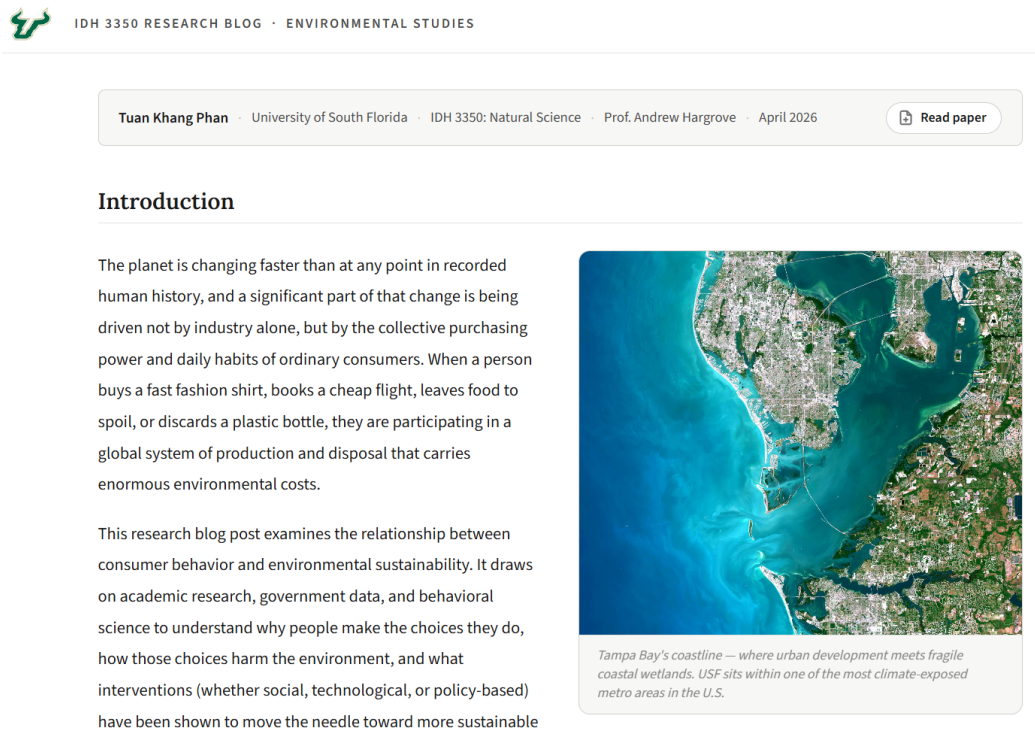


Figure 6. Consumer Behavior Survey interface (22 questions, 6 sections) and the post-submission results page showing verdict, per-section score bars, and full question breakdown.



IDH 3350 RESEARCH BLOG · ENVIRONMENTAL STUDIES

Tuan Khang Phan · University of South Florida · IDH 3350: Natural Science · Prof. Andrew Hargrove · April 2026 Read paper

Introduction

The planet is changing faster than at any point in recorded human history, and a significant part of that change is being driven not by industry alone, but by the collective purchasing power and daily habits of ordinary consumers. When a person buys a fast fashion shirt, books a cheap flight, leaves food to spoil, or discards a plastic bottle, they are participating in a global system of production and disposal that carries enormous environmental costs.

This research blog post examines the relationship between consumer behavior and environmental sustainability. It draws on academic research, government data, and behavioral science to understand why people make the choices they do, how those choices harm the environment, and what interventions (whether social, technological, or policy-based) have been shown to move the needle toward more sustainable

Tampa Bay's coastline — where urban development meets fragile coastal wetlands. USF sits within one of the most climate-exposed metro areas in the U.S.

Figure 7. Companion research blog publishing the IDH 3350 paper *"Consumer Behavior Impacts on Environmental Sustainability"* with abstract, literature review, and discussion sections.

B. Survey Instrument (Full Question List)

Scale: 1 = Strongly Disagree 2 = Moderately Disagree 3 = Neutral 4 = Moderately Agree 5 = Strongly Agree

Section A: Environmental Awareness

A1. I am aware of the environmental impact of the products I purchase daily.
A2. I regularly look for eco-labels or sustainability certifications when shopping.
A3. I understand what the terms "carbon footprint" and "circular economy" mean.
A4. I feel informed enough to make environmentally responsible purchasing decisions.

Section B: Attitudes Toward Sustainable Consumption

B1. Buying sustainably made products is important to me personally.
B2. I believe individual purchasing choices can make a meaningful difference to the environment.
B3. I would pay a premium for a product that is certified as environmentally friendly.
B4. I prefer buying secondhand or refurbished products over buying new when possible.

Section C: Actual Purchasing Behavior

C1. In the past month, I consciously avoided buying a product due to environmental concerns.

C2. I actively reduce the amount of single-use plastics I purchase.
C3. I research a brand's environmental practices before making a significant purchase.
C4. I choose local or low-packaging options at grocery stores or markets.

Section D: Social & Peer Influence

D1. My friends and peers influence my decision to buy eco-friendly products.
D2. I have shared information about sustainable products or practices on social media.
D3. I feel social pressure to appear environmentally conscious in my choices.

Section E: Barriers to Sustainable Shopping

E1. Price is the biggest barrier preventing me from buying more eco-friendly products.
E2. It is difficult to find trustworthy information about a product's sustainability.
E3. I find it inconvenient to shop sustainably compared to conventional shopping.

Section F: GreenCart Platform Impact

F1. A platform like GreenCart that scores product sustainability would change how I shop.
F2. Seeing a sustainability score on a product page would make me more likely to choose an eco-friendly option.
F3. Access to greener alternatives suggested by AI would help me make better purchasing decisions.
F4. Community features (volunteer matching, social feed) motivate me to take environmental action.